

If X and Y are CW complexes, then there is a natural CW structure on $X * Y$ having the subspaces X and Y as subcomplexes, with the remaining cells being the product cells of $X \times Y \times (0, 1)$. As usual with products, the CW topology on $X * Y$ may be weaker than the quotient of the product topology on $X \times Y \times I$.

Wedge Sum. This is a rather trivial but still quite useful operation. Given spaces X and Y with chosen points $x_0 \in X$ and $y_0 \in Y$, then the **wedge sum** $X \vee Y$ is the quotient of the disjoint union $X \amalg Y$ obtained by identifying x_0 and y_0 to a single point. For example, $S^1 \vee S^1$ is homeomorphic to the figure ‘8,’ two circles touching at a point. More generally one could form the wedge sum $\bigvee_{\alpha} X_{\alpha}$ of an arbitrary collection of spaces X_{α} by starting with the disjoint union $\bigsqcup_{\alpha} X_{\alpha}$ and identifying points $x_{\alpha} \in X_{\alpha}$ to a single point. In case the spaces X_{α} are cell complexes and the points x_{α} are 0-cells, then $\bigvee_{\alpha} X_{\alpha}$ is a cell complex since it is obtained from the cell complex $\bigsqcup_{\alpha} X_{\alpha}$ by collapsing a subcomplex to a point.

For any cell complex X , the quotient X^n/X^{n-1} is a wedge sum of n -spheres $\bigvee_{\alpha} S_{\alpha}^n$, with one sphere for each n -cell of X .

Smash Product. Like suspension, this is another construction whose importance becomes evident only later. Inside a product space $X \times Y$ there are copies of X and Y , namely $X \times \{y_0\}$ and $\{x_0\} \times Y$ for points $x_0 \in X$ and $y_0 \in Y$. These two copies of X and Y in $X \times Y$ intersect only at the point (x_0, y_0) , so their union can be identified with the wedge sum $X \vee Y$. The **smash product** $X \wedge Y$ is then defined to be the quotient $X \times Y / X \vee Y$. One can think of $X \wedge Y$ as a reduced version of $X \times Y$ obtained by collapsing away the parts that are not genuinely a product, the separate factors X and Y .

The smash product $X \wedge Y$ is a cell complex if X and Y are cell complexes with x_0 and y_0 0-cells, assuming that we give $X \times Y$ the cell-complex topology rather than the product topology in cases when these two topologies differ. For example, $S^m \wedge S^n$ has a cell structure with just two cells, of dimensions 0 and $m+n$, hence $S^m \wedge S^n = S^{m+n}$. In particular, when $m = n = 1$ we see that collapsing longitude and meridian circles of a torus to a point produces a 2-sphere.

Two Criteria for Homotopy Equivalence

Earlier in this chapter the main tool we used for constructing homotopy equivalences was the fact that a mapping cylinder deformation retracts onto its ‘target’ end. By repeated application of this fact one can often produce homotopy equivalences between rather different-looking spaces. However, this process can be a bit cumbersome in practice, so it is useful to have other techniques available as well. We will describe two commonly used methods here. The first involves collapsing certain subspaces to points, and the second involves varying the way in which the parts of a space are put together.

Collapsing Subspaces

The operation of collapsing a subspace to a point usually has a drastic effect on homotopy type, but one might hope that if the subspace being collapsed already has the homotopy type of a point, then collapsing it to a point might not change the homotopy type of the whole space. Here is a positive result in this direction:

|| *If (X, A) is a CW pair consisting of a CW complex X and a contractible subcomplex A , then the quotient map $X \rightarrow X/A$ is a homotopy equivalence.*

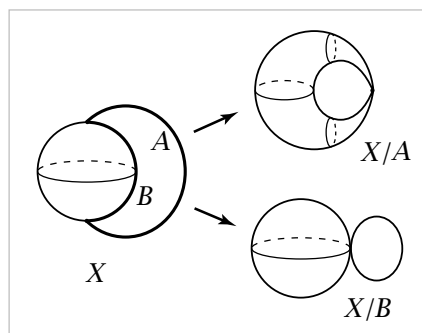
A proof will be given later in Proposition 0.17, but for now let us look at some examples showing how this result can be applied.

Example 0.7: Graphs. The three graphs ∞ ∞ ∞ are homotopy equivalent since each is a deformation retract of a disk with two holes, but we can also deduce this from the collapsing criterion above since collapsing the middle edge of the first and third graphs produces the second graph.

More generally, suppose X is any graph with finitely many vertices and edges. If the two endpoints of any edge of X are distinct, we can collapse this edge to a point, producing a homotopy equivalent graph with one fewer edge. This simplification can be repeated until all edges of X are loops, and then each component of X is either an isolated vertex or a wedge sum of circles.

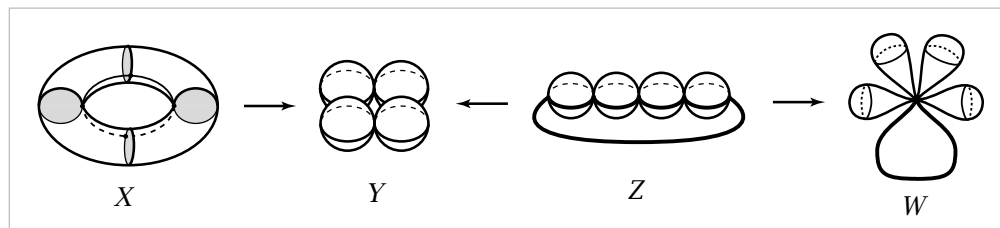
This raises the question of whether two such graphs, having only one vertex in each component, can be homotopy equivalent if they are not in fact just isomorphic graphs. Exercise 12 at the end of the chapter reduces the question to the case of connected graphs. Then the task is to prove that a wedge sum $\bigvee_m S^1$ of m circles is not homotopy equivalent to $\bigvee_n S^1$ if $m \neq n$. This sort of thing is hard to do directly. What one would like is some sort of algebraic object associated to spaces, depending only on their homotopy type, and taking different values for $\bigvee_m S^1$ and $\bigvee_n S^1$ if $m \neq n$. In fact the Euler characteristic does this since $\bigvee_m S^1$ has Euler characteristic $1 - m$. But it is a rather nontrivial theorem that the Euler characteristic of a space depends only on its homotopy type. A different algebraic invariant that works equally well for graphs, and whose rigorous development requires less effort than the Euler characteristic, is the fundamental group of a space, the subject of Chapter 1.

Example 0.8. Consider the space X obtained from S^2 by attaching the two ends of an arc A to two distinct points on the sphere, say the north and south poles. Let B be an arc in S^2 joining the two points where A attaches. Then X can be given a CW complex structure with the two endpoints of A and B as 0-cells, the interiors of A and B as 1-cells, and the rest of S^2 as a 2-cell. Since A and B are contractible,



X/A and X/B are homotopy equivalent to X . The space X/A is the quotient S^2/S^0 , the sphere with two points identified, and X/B is $S^1 \vee S^2$. Hence S^2/S^0 and $S^1 \vee S^2$ are homotopy equivalent, a fact which may not be entirely obvious at first glance.

Example 0.9. Let X be the union of a torus with n meridional disks. To obtain a CW structure on X , choose a longitudinal circle in the torus, intersecting each of the meridional disks in one point. These intersection points are then the 0-cells, the 1-cells are the rest of the longitudinal circle and the boundary circles of the meridional disks, and the 2-cells are the remaining regions of the torus and the interiors of the meridional disks. Collapsing each meridional disk to a point yields a homotopy



equivalent space Y consisting of n 2-spheres, each tangent to its two neighbors, a ‘necklace with n beads.’ The third space Z in the figure, a strand of n beads with a string joining its two ends, collapses to Y by collapsing the string to a point, so this collapse is a homotopy equivalence. Finally, by collapsing the arc in Z formed by the front halves of the equators of the n beads, we obtain the fourth space W , a wedge sum of S^1 with n 2-spheres. (One can see why a wedge sum is sometimes called a ‘bouquet’ in the older literature.)

Example 0.10: Reduced Suspension. Let X be a CW complex and $x_0 \in X$ a 0-cell. Inside the suspension SX we have the line segment $\{x_0\} \times I$, and collapsing this to a point yields a space ΣX homotopy equivalent to SX , called the **reduced suspension** of X . For example, if we take X to be $S^1 \vee S^1$ with x_0 the intersection point of the two circles, then the ordinary suspension SX is the union of two spheres intersecting along the arc $\{x_0\} \times I$, so the reduced suspension ΣX is $S^2 \vee S^2$, a slightly simpler space. More generally we have $\Sigma(X \vee Y) = \Sigma X \vee \Sigma Y$ for arbitrary CW complexes X and Y . Another way in which the reduced suspension ΣX is slightly simpler than SX is in its CW structure. In SX there are two 0-cells (the two suspension points) and an $(n+1)$ -cell $e^n \times (0, 1)$ for each n -cell e^n of X , whereas in ΣX there is a single 0-cell and an $(n+1)$ -cell for each n -cell of X other than the 0-cell x_0 .

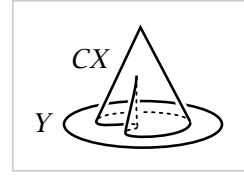
The reduced suspension ΣX is actually the same as the smash product $X \wedge S^1$ since both spaces are the quotient of $X \times I$ with $X \times \partial I \cup \{x_0\} \times I$ collapsed to a point.

Attaching Spaces

Another common way to change a space without changing its homotopy type involves the idea of continuously varying how its parts are attached together. A general definition of ‘attaching one space to another’ that includes the case of attaching cells

is the following. We start with a space X_0 and another space X_1 that we wish to attach to X_0 by identifying the points in a subspace $A \subset X_1$ with points of X_0 . The data needed to do this is a map $f: A \rightarrow X_0$, for then we can form a quotient space of $X_0 \amalg X_1$ by identifying each point $a \in A$ with its image $f(a) \in X_0$. Let us denote this quotient space by $X_0 \sqcup_f X_1$, the space X_0 with X_1 **attached along A via f** . When $(X_1, A) = (D^n, S^{n-1})$ we have the case of attaching an n -cell to X_0 via a map $f: S^{n-1} \rightarrow X_0$.

Mapping cylinders are examples of this construction, since the mapping cylinder M_f of a map $f: X \rightarrow Y$ is the space obtained from Y by attaching $X \times I$ along $X \times \{1\}$ via f . Closely related to the mapping cylinder M_f is the **mapping cone** $C_f = Y \sqcup_f CX$ where CX is the cone $(X \times I)/(X \times \{0\})$ and we attach this to Y along $X \times \{1\}$ via the identifications $(x, 1) \sim f(x)$. For example, when X is a sphere S^{n-1} the mapping cone C_f is the space obtained from Y by attaching an n -cell via $f: S^{n-1} \rightarrow Y$. A mapping cone C_f can also be viewed as the quotient M_f/X of the mapping cylinder M_f with the subspace $X = X \times \{0\}$ collapsed to a point.

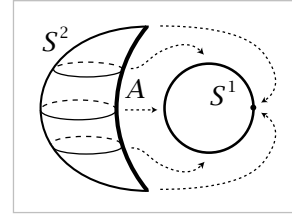


If one varies an attaching map f by a homotopy f_t , one gets a family of spaces whose shape is undergoing a continuous change, it would seem, and one might expect these spaces all to have the same homotopy type. This is often the case:

|| If (X_1, A) is a CW pair and the two attaching maps $f, g: A \rightarrow X_0$ are homotopic, then $X_0 \sqcup_f X_1 \simeq X_0 \sqcup_g X_1$.

Again let us defer the proof and look at some examples.

Example 0.11. Let us rederive the result in Example 0.8 that a sphere with two points identified is homotopy equivalent to $S^1 \vee S^2$. The sphere with two points identified can be obtained by attaching S^2 to S^1 by a map that wraps a closed arc A in S^2 around S^1 , as shown in the figure. Since A is contractible, this attaching map is homotopic to a constant map, and attaching S^2 to S^1 via a constant map of A yields $S^1 \vee S^2$. The result then follows since (S^2, A) is a CW pair, S^2 being obtained from A by attaching a 2-cell.



Example 0.12. In similar fashion we can see that the necklace in Example 0.9 is homotopy equivalent to the wedge sum of a circle with n 2-spheres. The necklace can be obtained from a circle by attaching n 2-spheres along arcs, so the necklace is homotopy equivalent to the space obtained by attaching n 2-spheres to a circle at points. Then we can slide these attaching points around the circle until they all coincide, producing the wedge sum.

Example 0.13. Here is an application of the earlier fact that collapsing a contractible subcomplex is a homotopy equivalence: If (X, A) is a CW pair, consisting of a cell

complex X and a subcomplex A , then $X/A \simeq X \cup CA$, the mapping cone of the inclusion $A \hookrightarrow X$. For we have $X/A = (X \cup CA)/CA \simeq X \cup CA$ since CA is a contractible subcomplex of $X \cup CA$.

Example 0.14. If (X, A) is a CW pair and A is contractible in X , that is, the inclusion $A \hookrightarrow X$ is homotopic to a constant map, then $X/A \simeq X \vee SA$. Namely, by the previous example we have $X/A \simeq X \cup CA$, and then since A is contractible in X , the mapping cone $X \cup CA$ of the inclusion $A \hookrightarrow X$ is homotopy equivalent to the mapping cone of a constant map, which is $X \vee SA$. For example, $S^n/S^i \simeq S^n \vee S^{i+1}$ for $i < n$, since S^i is contractible in S^n if $i < n$. In particular this gives $S^2/S^0 \simeq S^2 \vee S^1$, which is Example 0.8 again.

The Homotopy Extension Property

In this final section of the chapter we will actually prove a few things. In particular we prove the two criteria for homotopy equivalence described above, along with the fact that any two homotopy equivalent spaces can be embedded as deformation retracts of the same space.

The proofs depend upon a technical property that arises in many other contexts as well. Consider the following problem. Suppose one is given a map $f_0: X \rightarrow Y$, and on a subspace $A \subset X$ one is also given a homotopy $f_t: A \rightarrow Y$ of $f_0|_A$ that one would like to extend to a homotopy $f_t: X \rightarrow Y$ of the given f_0 . If the pair (X, A) is such that this extension problem can always be solved, one says that (X, A) has the **homotopy extension property**. Thus (X, A) has the homotopy extension property if every map $X \times \{0\} \cup A \times I \rightarrow Y$ can be extended to a map $X \times I \rightarrow Y$.

In particular, the homotopy extension property for (X, A) implies that the identity map $X \times \{0\} \cup A \times I \rightarrow X \times \{0\} \cup A \times I$ extends to a map $X \times I \rightarrow X \times \{0\} \cup A \times I$, so $X \times \{0\} \cup A \times I$ is a retract of $X \times I$. The converse is also true: If there is a retraction $X \times I \rightarrow X \times \{0\} \cup A \times I$, then by composing with this retraction we can extend every map $X \times \{0\} \cup A \times I \rightarrow Y$ to a map $X \times I \rightarrow Y$. Thus the homotopy extension property for (X, A) is equivalent to $X \times \{0\} \cup A \times I$ being a retract of $X \times I$. This implies for example that if (X, A) has the homotopy extension property, then so does $(X \times Z, A \times Z)$ for any space Z , a fact that would not be so easy to prove directly from the definition.

If (X, A) has the homotopy extension property, then A must be a closed subspace of X , at least when X is Hausdorff. For if $r: X \times I \rightarrow X \times \{0\} \cup A \times I$ is a retraction onto the subspace $X \times \{0\} \cup A \times I$, then the image of r is the set of points $z \in X \times I$ with $r(z) = z$, a closed set if X is Hausdorff, so $X \times \{0\} \cup A \times I$ is closed in $X \times I$ and hence A is closed in X .

A simple example of a pair (X, A) with A closed for which the homotopy extension property fails is the pair (I, A) where $A = \{0, 1, 1/2, 1/3, 1/4, \dots\}$. It is not hard to show that there is no continuous retraction $I \times I \rightarrow I \times \{0\} \cup A \times I$. The breakdown of homotopy extension here can be attributed to the bad structure of (X, A) near 0.